A positive answer to the height ω_2 Valdivia r-tree problem

Run `fa_banach_001`; source arXiv:2305.11737

Source problem

Russo and Somaglia's paper *Banach spaces of continuous functions without norming Markushevich bases* proves that every scattered compact r-tree of height strictly less than ω_2 is Valdivia. Immediately after that result they ask the boundary question below.

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Problem 5.9. *Let T be a scattered, compact r-tree with* $\mathrm{ht}(T) = \omega_2$*. Must T be Valdivia?*

In the light of the previous results, it is natural to wonder when compact trees are scattered. Therefore, we now proceed to give a characterisation of scattered trees. Given

Figure 1: Russo–Somaglia, arXiv:2305.11737, page 15: Problem 5.9.

Problem 5.9 asks:

> Let T be a scattered compact r-tree with $\mathrm{ht}(T) = \omega_2$. Must T be Valdivia?

Definitions and cited facts

All trees are endowed with the coarse wedge topology. For $t \in T$,

$$V_t = \{u \in T : t \leq u\}, \qquad \mathrm{ims}(t) = \{u \in T : t < u,\ \mathrm{ht}(u, T) = \mathrm{ht}(t, T) + 1\}.$$

A tree is an r-tree if $|\,\mathrm{ims}(t)| < \omega$ whenever $\mathrm{cf}(t) \geq \omega_1$. Put

$$D(T) = \{t \in T : \mathrm{cf}(t) \leq \omega\}.$$

We use three standard facts from the source and the cited compact-tree literature.

- Russo–Somaglia Corollary 5.7: every scattered compact r-tree of height $< \omega_2$ is Valdivia [1].
- Russo–Somaglia Lemma 5.5: a scattered compact tree of height $\leq \omega_2$ can be decomposed at its top Cantor–Bendixson derivative into pairwise disjoint clopen, $\wedge$-closed, lower-rank subtrees and a finite union of chains of cardinality at most ω_1 [1].
- Somaglia/Kalenda clopen-family criterion: a compact zero-dimensional space, in particular a compact tree in the coarse wedge topology, is Valdivia provided it has a dense set D and a T_0-separating family of clopen sets that is point-countable on D. For compact trees one may take $D = D(T)$ [2].

Proof intuition

The source paper's lower-height theorem fails to apply directly at ω_2 because the known height induction reaches only levels below ω_2. The boundary case becomes accessible if one inducts instead on Cantor–Bendixson rank.

Take a hypothetical counterexample of minimal Cantor–Bendixson rank. The source paper's scattered-tree decomposition cuts off all top-rank behavior along finitely many chains. Everything off those chains is a clopen subtree of smaller Cantor–Bendixson rank, hence Valdivia by minimality, even if some of those pieces still have height ω_2. Collapse each such clopen piece to a marker node. The quotient is only a finite-spine tree, so its height is strictly below ω_2, and it is Valdivia by Russo–Somaglia's already-proved theorem.

The remaining point is a replacement lemma: a Valdivia zero-dimensional quotient remains Valdivia after replacing marker nodes, lying in the dense Σ-part, by clopen Valdivia compact fibers. Pull back a point-countable clopen separating family from the quotient and add the point-countable separating families inside the fibers.

A clopen replacement lemma

Lemma 1 (Clopen replacement)**.** *Let K and L be compact zero-dimensional spaces and let $q : K \to L$ be a continuous surjection. Suppose that L is Valdivia, witnessed by a dense set $E \subset L$ and a T_0-separating family $\mathcal{A}$ of clopen subsets of L point-countable on E.*

Assume there is a set $M \subset E$ such that:

1. *if $y \notin M$, then $q^{-1}(y)$ is a singleton;*
2. *if $m \in M$, then $K_m = q^{-1}(m)$ is a clopen Valdivia compactum;*
3. *for every $m \in M$, K_m has a dense set E_m and a T_0-separating clopen family $\mathcal{A}_m$, taken inside K_m, that is point-countable on E_m.*

Finally assume that

$$E_K = q^{-1}(E \setminus M) \cup \bigcup_{m \in M} E_m$$

is dense in K. Then K is Valdivia.

Proof. For $A \in \mathcal{A}$, the preimage $q^{-1}(A)$ is clopen in K. Since each K_m is clopen, every member of every internal family $\mathcal{A}_m$ is also clopen in K. Define

$$\mathcal{B} = \{q^{-1}(A) : A \in \mathcal{A}\} \cup \bigcup_{m \in M} \mathcal{A}_m$$

and

$$E_K = q^{-1}(E \setminus M) \cup \bigcup_{m \in M} E_m.$$

The family $\mathcal{B}$ is T_0-separating. If two points have different images in L, a member of $\mathcal{A}$ separates their images, and its preimage separates the points. If they have the same image, then they lie in one fiber K_m, and a member of $\mathcal{A}_m$ separates them.

Finally, $\mathcal{B}$ is point-countable on E_K. If $x \in q^{-1}(E \setminus M)$, the only members of $\mathcal{B}$ containing x are pullbacks of members of $\mathcal{A}$ containing $q(x)$, a countable family. If $x \in E_m$, then the pullback part contributes only the countably many $A \in \mathcal{A}$ with $m \in A$, and the internal part contributes only the countably many members of $\mathcal{A}_m$ containing x. Therefore K has a dense set with a point-countable T_0-separating clopen family, so K is Valdivia by the criterion. $\square$

The finite-spine quotient

Lemma 2 (Finite-spine quotient)**.** *Let T be a scattered compact r-tree with* $\mathrm{ht}(T) \le \omega_2$ *and Cantor–Bendixson rank $\rho + 1$. There is a continuous surjection*

$$q : T \longrightarrow Q$$

onto a scattered compact r-tree Q of height $< \omega_2$ such that:

1. *every non-singleton fiber of q is a clopen, $\wedge$-closed, compact subtree of T of Cantor–Bendixson rank $< \rho + 1$;*
2. *the points of Q corresponding to non-singleton fibers are on successor levels, or are the root of Q, hence belong to $D(Q)$;*
3. *singleton fibers are exactly the points lying on a finite union of chains of T.*

Proof. Russo–Somaglia Lemma 5.5 gives the required decomposition. We recall the part of that construction that matters here. Since T is compact and scattered, $T^{(\rho)}$ is finite. Around each $t \in T^{(\rho)}$ choose a clopen wedge neighborhood $W^{F_t}_{s_0(t)}$ isolating t from the other top-rank points. The proof of Lemma 5.5 then writes T as a pairwise disjoint union of:

- a clopen $\wedge$-closed remainder
$$S = T \setminus \bigcup_{t \in T^{(\rho)}} V_{s_0(t)}$$
with rank $< \rho + 1$;
- clopen $\wedge$-closed side pieces V_r occurring along the finite chains from $s_0(t)$ to t, each of rank $< \rho + 1$;
- clopen $\wedge$-closed pieces $W^{G_r}_r$ immediately above the top-rank points, again each of rank $< \rho+1$;
- the finite union of residual chains
$$C = \bigcup_{t \in T^{(\rho)}} \big(\hat{t} \cap V_{s_0(t)}\big).$$

The rank inequalities are exactly the point of the decomposition in Russo–Somaglia's proof of Theorem 5.6.

Collapse every clopen lower-rank piece in the first three classes to one marker point and leave the residual chain points distinct. The order on the quotient Q is the evident tree order: a marker replacing a side piece takes the position of the root of that side piece; the marker replacing S, if $S \ne \emptyset$, is the root below the finitely many residual spines. Because the residual part is a finite union of chains and every point of T has height $< \omega_2$, the quotient has height $< \omega_2$. Adding marker successors cannot reach height ω_2.

The quotient is chain complete and rooted: every chain either is eventually inside one residual chain, or terminates at a marker replacing one collapsed clopen subtree. The r-tree condition is inherited from T. At a quotient point of cofinality at least ω_1, its immediate successors come from immediate successors of the corresponding point of T, and there are finitely many because T is an r-tree. The quotient is scattered because it is a finite union of ordinal chains with isolated marker successors.

Finally, q is continuous. The basic clopen subsets of Q are generated by wedges V_a, where a is the root or lies on a successor level, and their complements. If a is a marker, $q^{-1}(V_a)$ is precisely the corresponding clopen piece of T. If a lies on a residual chain, then $q^{-1}(V_a)$ is the corresponding wedge in T, possibly with finitely many already clopen collapsed side pieces removed or added according to the finite Boolean description of the basic wedge. Hence each basic clopen of Q has clopen preimage in T. This proves the asserted properties. □

Main theorem

Theorem 1. *Let T be a scattered compact r-tree with* $\mathrm{ht}(T) = \omega_2$*. Then T is Valdivia.*

Proof. Assume, toward a contradiction, that there is a non-Valdivia scattered compact r-tree of height ω_2. Choose such a tree T with minimal Cantor–Bendixson rank.

Apply Lemma 2. For every non-singleton fiber $P = q^{-1}(m)$, the set P is a clopen, $\wedge$-closed, compact subtree of T, and its Cantor–Bendixson rank is strictly smaller than the rank of T. It is again scattered and an r-tree, with height at most ω_2. If $\mathrm{ht}(P) < \omega_2$, then P is Valdivia by Russo–Somaglia Corollary 5.7. If $\mathrm{ht}(P) = \omega_2$, then P is Valdivia by the minimality of T's Cantor–Bendixson rank. Thus every non-singleton fiber of q is Valdivia.

The quotient Q is a scattered compact r-tree of height $< \omega_2$, so Q is Valdivia by Russo–Somaglia Corollary 5.7. Moreover, the marker points corresponding to non-singleton fibers are in $D(Q)$, by Lemma 2. For each non-singleton fiber P, choose a dense Σ-set and a point-countable clopen separating family witnessing that P is Valdivia. For Q, choose the corresponding witness with $D(Q)$ as the dense set, as supplied by the compact-tree criterion. The dense set required in Lemma 1 is dense in T: inside each clopen fiber this is the density of its chosen dense Σ-set, and along the residual finite spine it is the usual density of points of countable cofinality in a compact tree.

All hypotheses of the clopen replacement Lemma 1 are now satisfied. Therefore T is Valdivia, contradicting the choice of T. This proves the theorem. □

Relation to the earlier partial packet

The earlier root-split packet handled the special case where every immediate successor subtree above the root has height $< \omega_2$. The proof above removes that extra hypothesis. Subtrees of height ω_2 may occur below the root, but once the tree is split by Cantor–Bendixson rank, those tall subtrees have lower rank and are handled by the minimal-rank induction.

Verification and novelty notes

The proof is non-computational. The local verification script checks that the packet includes the source problem, the full theorem, the replacement lemma, and the minimal-rank argument.

Local cheap-index searches covered the arXiv id `2305.11737`, source title, `Problem 5.9`, `scattered compact r-tree`, `height omega_2`, `Valdivia compact tree`, and nearby norming-M-basis phrases. The only existing packet for this arXiv id was the root-split partial subcase produced earlier in this run.

External web searches on June 16, 2026 used the exact source title, arXiv id, `Can the first statement of Theorem 3.2 be reversed`, and variants of `compact r-tree height omega_2 Valdivia`. They found the source paper and Somaglia's cited compact-tree characterization, but no later paper explicitly resolving Problem 5.9.

Human review recommendation

Human review should focus first on Lemma 2. In particular, check that the quotient obtained from the Russo–Somaglia Lemma 5.5 decomposition is indeed a compact r-tree of height $< \omega_2$, and that the quotient map pulls back basic clopen sets to clopen sets of T. Once that finite-spine quotient is accepted, Lemma 1 and the minimal Cantor–Bendixson rank contradiction are routine.

References

[1] T. Russo and J. Somaglia, *Banach spaces of continuous functions without norming Markushevich bases*, arXiv:2305.11737.

[2] J. Somaglia, *On compact trees with the coarse wedge topology*, arXiv:1803.11107.

[3] O. F. K. Kalenda, *Valdivia compact spaces in topology and Banach space theory*, Extracta Math. 15 (2000), 1–85.